NAME: _____ Geometry (FL B.E.S.T.)

2.4 Parallel & Perpendicular Equations

Slope connects the visual idea of parallel and perpendicular lines to hard algebra: equal slopes force lines to run in the same direction forever without meeting, while a slope and its negative reciprocal force a perfect $90°$ turn. This section goes beyond just flipping a fraction — it uses the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ from two raw points, tests whether three points are collinear, and solves for a missing coordinate that makes two lines parallel or perpendicular.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Slope of line parallel to slope 7?

2. Perpendicular to slope 5?

3. Perpendicular to slope $-\frac{2}{3}$?

4. Parallel to $y = -x + 4$ through $(0, 1)$.

5. Perpendicular to $y = \frac{1}{2}x$ through $(0, -3)$.

6. $y = 3x + 2$ and $y = 3x - 5$ parallel?

7. $y = 2x$ and $y = 2x + 1$ perpendicular?

8. $y = -\frac{1}{4}x$ and $y = 4x + 3$ perpendicular?

9. Slope of $y = 5$?

10. Equation of vertical line through $x = 3$?

11. Line perpendicular to $y = 2$?

12. If lines are perpendicular (neither vertical), product of slopes?

13. Parallel to $y = 4x - 1$ through $(0, 0)$.

14. Determine whether $A(-2, -1)$, $B(1, 5)$, and $C(3, 9)$ are collinear, showing the slope work.

15. A line through $(1, 4)$ and $(m, 10)$ is parallel to $y = \frac{3}{2}x - 1$. Find m.

✓ Answer Key

1. 7
2. $-\frac{1}{5}$
3. $\frac{3}{2}$
4. $y = -x + 1$
5. $y = -2x - 3$
6. Yes
7. No
8. Yes
9. 0
10. $x = 3$
11. A vertical line
12. -1
13. $y = 4x$
14. Yes; slope $AB = 2 =$ slope BC
15. $m = 5$